Java Data & Control Flow Analyzer — Screenshots & Reflection

By: Ramyah Murphy

```
PROBLEMS 4    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

Enter vehicle fuel efficiency (mpg): 30
Enter price per gallon: 3.50
Enter number of passengers: 2

--- Trip Cost Summary ---
Gallons Needed: 4.00
Total Fuel Cost: $14.00
Cost Per Mile: $0.12
Cost Per Passenger: $7.00
PS C:\Users\ramya>
```

```
Enter retirement contribution percentage: 10
Print detailed pay stub? (Y/N): y

--- Weekly Pay Stub for tod ---
Regular Hours: 40.00
Overtime Hours: 40.00
Gross Pay: $1500.00
Retirement Deduction: $150.00
Net Pay: $1350.00
PS C:\Users\ramya>
```

Working through these two Java exercises helped me understand how important it is to use the correct data types, operators, and control flow when building programs. One challenge I faced was making sure the user inputs were validated properly. If I forgot to check for negative values or type something incorrectly, the program would crash or give the wrong output. Another challenge was formatting the results, so they were clear and easy to read, especially when using decimals and percentages.

Choosing the right data types matters because it affects accuracy and prevents errors. For example, using `double` fuel calculations and pay computations avoids rounding issues, while using `int` for passengers and percentages prevents invalid fractional values. I also learned how type conversion, like dividing by `100.0`, ensures the math works correctly.

A real-world application of Exercise 1 would be a trip-planning or rideshare cost estimator. Exercise 2 relates directly to payroll systems that calculate wages, overtime, and deductions for employees